

smallantimagmas

**A library of antiassociative magmas of
small order**

0.6.0

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Chapter 1

smallantimagmas automatic generated documentation

1.1 smallantimagmas automatic generated documentation of properties

1.1.1 IsAntiassociative (for IsMagma)

▷ `IsAntiassociative( $M$ )` (property)
Returns: true or false
identifies whether magma M is antiassociative.

Example

```
gap> IsAntiassociative(OneSmallGroup(16));  
false  
gap> IsAntiassociative(OneSmallAntimagma(2));  
true  
gap> IsAntiassociative(OneSmallAntimagma(3));  
true
```

1.1.2 IsLeftCyclic (for IsMagma)

▷ `IsLeftCyclic( $M$ )` (property)
Returns: true or false
if magma is left cyclic m .

1.1.3 IsRightCyclic (for IsMagma)

▷ `IsRightCyclic( $M$ )` (property)
Returns: true or false
if magma is left cyclic m .

1.1.4 IsLeftDistributive (for IsMagma)

▷ `IsLeftDistributive( $M$ )` (property)
Returns: true or false
if magma is left distributive m .

Example

```
gap> List(AllSmallAntimagmas(3), M -> IsLeftDistributive(M));
[ true, false, false, false, false ]
```

1.1.5 IsRightDistributive (for IsMagma)

▷ IsRightDistributive(M) (property)

Returns: true or false
if magma is right distributive m .

Example

```
gap> List(AllSmallAntimagmas(3), M -> IsRightDistributive(M));
[ false, false, false, false, true ]
```

1.1.6 IsLeftCancellative (for IsMagma)

▷ IsLeftCancellative(M) (property)

Returns: true or false
if magma is left cancellative m .

Example

```
gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> Display(MultiplicationTable(M));
[ [ 2, 1 ],
  [ 2, 1 ] ]
gap> IsRightCancellative(M);
false
gap> IsLeftCancellative(M);
true
gap> List(AllSmallAntimagmas(2), M -> IsLeftCancellative(M));
[ true ]
```

1.1.7 IsRightCancellative (for IsMagma)

▷ IsRightCancellative(M) (property)

Returns: true or false
if magma is right cancellative m .

Example

```
gap> List(AllSmallAntimagmas(2), M -> IsRightCancellative(M));
[ false ]
```

1.1.8 IsCancellative (for IsMagma)

▷ IsCancellative(M) (property)

Returns: true or false
if magma is cancellative m .

Example

```
gap> List(AllSmallAntimagmas(2), M -> IsCancellative(M));
[ false ]
```

1.1.9 IsLeftFPFInducted (for IsMagma)

▷ `IsLeftFPFInducted( $M$ )` (property)

Returns: true or false

is a left-hand sided fixed-point free inducted m .

Example

```
gap> Display(MultiplicationTable(TransposedMagma(SmallAntimagma(2, 1))));
[ [ 2, 2 ],
  [ 1, 1 ] ]
gap> IsLeftFPFInducted(TransposedMagma(SmallAntimagma(2, 1)));
true
```

1.1.10 IsRightFPFInducted (for IsMagma)

▷ `IsRightFPFInducted( $M$ )` (property)

Returns: true or false

is a right-hand sided fixed-point free inducted m .

Example

```
gap> Display(MultiplicationTable(SmallAntimagma(2, 1)));
[ [ 2, 1 ],
  [ 2, 1 ] ]
gap> IsRightFPFInducted(SmallAntimagma(2, 1));
true
```

1.1.11 IsLeftDerangementInducted (for IsMagma)

▷ `IsLeftDerangementInducted( $M$ )` (property)

Returns: true or false

is a left-hand sided derangement inducted m .

Example

```
gap> M := TransposedMagma(SmallAntimagma(2, 1));
<magma with 2 generators>
gap> IsLeftFPFInducted(M);
true
gap> IsRightFPFInducted(M);
false
gap> IsRightDerangementInducted(M);
false
```

1.1.12 IsRightDerangementInducted (for IsMagma)

▷ `IsRightDerangementInducted( $M$ )` (property)

Returns: true or false

is a right-hand sided derangement inducted m .

Example

```
gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> IsLeftFPFInducted(M);
false
```

```
gap> IsRightFPFInducted(M);
true
gap> IsRightDerangementInducted(M);
true
```

1.1.13 IsLeftAlternative (for IsMagma)

- ▷ IsLeftAlternative(M) (property)
Returns: true or false
 is a left-alternative magma M .

Example

1.1.14 IsRightAlternative (for IsMagma)

- ▷ IsRightAlternative(M) (property)
Returns: true or false
 is a right-alternative magma M .

Example

1.2 smallantimagmas automatic generated documentation of attributes

1.2.1 AssociativityIndex (for IsMagma)

- ▷ AssociativityIndex(M) (attribute)
 identifies associativity index of M .

Example

```
gap> OneSmallAntimagma(2);
<magma with 2 generators>
gap> AssociativityIndex(OneSmallAntimagma(2));
0
gap> OneSmallGroup(4);
<pc group of size 4 with 2 generators>
gap> AssociativityIndex(OneSmallGroup(4));
64
gap> AssociativityIndex(OneSmallGroup(4)) = 4 ^ 3;
true
```

1.2.2 DiagonalOfMultiplicationTable (for IsMagma)

- ▷ DiagonalOfMultiplicationTable(M) (attribute)
 computes diagonal of multiplication table of M .

Example

```
gap> List(AllSmallAntimagmas(3), M -> DiagonalOfMultiplicationTable((M)));
[ [ 2, 1, 1 ], [ 2, 1, 1 ],
  [ 2, 3, 2 ], [ 2, 1, 1 ],
  [ 2, 3, 1 ]
]
```

1.2.3 CommutativityIndex (for IsMagma)

▷ `CommutativityIndex( $M$ )` (attribute)

identifies commutativity index of M .

Example

1.2.4 AnticommutativityIndex (for IsMagma)

▷ `AnticommutativityIndex( $M$ )` (attribute)

calculates anticommutativity index of M .

Example

1.2.5 SquaresIndex (for IsMagma)

▷ `SquaresIndex( $M$ )` (attribute)

computes squares index of M so the order of $\{m^2 | m \in M\}$.

Example

```
gap> List(AllSmallAntimagmas(2), M -> List(M, m -> m * m));
[ [ m2, m1 ] ]
gap> List(AllSmallAntimagmas(2), M -> SquaresIndex(M));
[ 2 ]
gap> List(AllSmallAntimagmas(3), M -> SquaresIndex(M));
[ 2, 2, 2, 2, 3 ]
```

1.2.6 IdSmallAntimagma (for IsMagma)

▷ `IdSmallAntimagma( $M$ )` (attribute)

identifies class of antiassociative magma M .

Example

```
gap> IsAntiassociative(OneSmallGroup(16));
false
gap> IsAntiassociative(OneSmallAntimagma(2));
true
gap> IsAntiassociative(OneSmallAntimagma(3));
true
```

1.2.7 LeftOrder (for IsExtLElement)

▷ `LeftOrder( $[m]$ )` (attribute)

returns a left order of element m .

1.2.8 RightOrder (for IsExtRElement)

▷ `RightOrder([m])` (attribute)

returns a right order of element m .

1.2.9 LeftOrdersOfElements (for IsMagma)

▷ `LeftOrdersOfElements([m])` (attribute)

returns a left order of element m .

1.2.10 RightOrdersOfElements (for IsMagma)

▷ `RightOrdersOfElements([m])` (attribute)

returns a left order of element m .

1.2.11 DigraphOfDiagonal (for IsMagma)

▷ `DigraphOfDiagonal(M)` (attribute)

builds a digraph from the diagonal of M .

Example

1.3 smallantimagnas automatic generated documentation of global functions

1.3.1 AllSubmagmas

▷ `AllSubmagmas(M)` (function)

builds a collection of non-isomorphic submagmas of M .

Example

```
gap> AllSmallAntimagnas(2);
[ <magma with 2 generators> ]
gap> List(AllSmallAntimagnas(2), M -> AllSubmagmas(M));
[ [ <magma with 1 generator> ] ]
```

1.3.2 MagmaIsomorphismInvariantsMatch

▷ `MagmaIsomorphismInvariantsMatch(M)` (function)

computes isomorphism invariants of M .

1.3.3 IsMagmaIsomorphic

▷ IsMagmaIsomorphic(M , N) (function)

identifies whether magmas M , N are isomorphic.

Example

```
gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> N := TransposedMagma(M);
<magma with 2 generators>
gap> T := MagmaByMultiplicationTable([[2, 1], [2, 1]]);
<magma with 2 generators>
gap> IsMagmaIsomorphic(M, M);
true
gap> IsMagmaIsomorphic(M, T);
true
gap> IsMagmaIsomorphic(M, N);
false
```

1.3.4 IsMagmaAntiisomorphic

▷ IsMagmaAntiisomorphic($[M, N]$) (function)

identifies whether magmas M , N are antiisomorphic.

Example

```
gap> N := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> N := TransposedMagma(M);
<magma with 2 generators>
gap> IsMagmaAntiisomorphic(M, M);
false
gap> IsMagmaAntiisomorphic(M, N);
true
gap> IsMagmaAntiisomorphic(M, TransposedMagma(M));
true
```

1.3.5 TransposedMagma

▷ TransposedMagma($[M]$) (function)

generates transposed magma M .

Example

```
gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> IsMagmaAntiisomorphic(M, TransposedMagma(M));
true
gap> IsMagmaIsomorphic(M, TransposedMagma(TransposedMagma(M)));
true
```

```

gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> Display(MultiplicationTable(M));
[ [ 2, 1 ],
  [ 2, 1 ] ]
gap> Display(MultiplicationTable(TransposedMagma(M)));
[ [ 2, 2 ],
  [ 1, 1 ] ]

```

1.3.6 LeftPower

▷ LeftPower($[m, k]$) (function)

returns a left k -power of element m .

1.3.7 RightPower

▷ RightPower($[m, k]$) (function)

returns a right k -power of element m .

1.3.8 AllSmallAntimagmas

▷ AllSmallAntimagmas(n) (function)

returns all antiassociative magmas of specified size n (a number), up to isomorphism and anti-isomorphism, i.e. keeping a single representative for every class of magmas that are pairwise isomorphic or anti-isomorphic.

Example

```

gap> AllSmallAntimagmas(2);
[ <magma with 2 generators> ]
gap> AllSmallAntimagmas(3);
[
  <magma with 3 generators>, <magma with 3 generators>, <magma with 3 generators>,
  <magma with 3 generators>, <magma with 3 generators>
]

```

1.3.9 NrSmallAntimagmas

▷ NrSmallAntimagmas(n) (function)

counts number of antiassociative magmas of specified size n (a number), up to isomorphism and anti-isomorphism.

Example

```

gap> NrSmallAntimagmas(2);
1
gap> NrSmallAntimagmas(3);
5

```

```
gap> NrSmallAntimagnas(4);
8891
```

1.3.10 SmallAntimagma

▷ `SmallAntimagma([n, i])` (function)

returns antiassociative magma of id $[n, i]$, up to isomorphism and anti-isomorphism.

Example

```
gap> SmallAntimagma(2, 1);
<magma with 2 generators>
gap> SmallAntimagma(4, 5);
<magma with 4 generators>
gap> SmallAntimagma([4, 5]);
<magma with 4 generators>
```

1.3.11 OneSmallAntimagma

▷ `OneSmallAntimagma(n)` (function)

returns a random antiassociative magma of size n , up to isomorphism and anti-isomorphism.

Example

```
gap> OneSmallAntimagma(2);
<magma with 2 generators>

gap> OneSmallAntimagma(3);
<magma with 3 generators>
```

1.4 smallantimagnas automatic generated documentation of methods

1.4.1 MagmaIsomorphism (for IsMagma, IsMagma)

▷ `MagmaIsomorphism(M, N)` (operation)

computes an isomorphism between magmas M, N .

Example

```
gap> M := SmallAntimagma(2, 1);
<magma with 2 generators>
gap> N := MagmaByMultiplicationTable([[2, 1], [2, 1]]);
<magma with 2 generators>
gap> MagmaIsomorphism(M, N);
<general mapping: Domain([ m1, m2 ]) -> Domain([ m1, m2 ]) >
```

1.4.2 MagmaAntiisomorphism (for IsMagma, IsMagma)

▷ `MagmaAntiisomorphism(M, N)` (operation)

creates an antiisomorphism between magmas M, N .

Example

```
gap> M := SmallAntimagma(2, 1);  
<magma with 2 generators>  
gap> N := TransposedMagma(M);  
<magma with 2 generators>  
gap> MagmaAntiisomorphism(M, N);  
<mapping: Domain([ m1, m2 ]) -> Domain([ m1, m2 ]) >
```

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